Reference:

[1] Augmentative and Alternative Communication (AAC): Overview

Authors: Beukelman & Mirenda

Source: Google Scholar

Link: https://brookespublishing.com/wp-content/uploads/2020/01/Beukelman-First-Proofs-Excerpt.pdf

Problem:

Many individuals with complex communication needs (CCN) cannot communicate effectively through speech or writing, leading to social, emotional, and educational difficulties.

Solution:

AAC (Augmentative and Alternative Communication) systems are used to support communication. These include:

- Unaided AAC: gestures, signs

- Aided AAC: communication boards, mobile apps, speech-generating devices

Working:

Users select symbols, icons, or words, which are converted into text or speech to communicate.

Key Concept:

Communication requires "communicative competence" including:

- Linguistic (language understanding)

- Operational (using the system)

- Social (interaction skills)

- Strategic (handling communication breakdowns)

Limitations:

- Devices alone are not enough (need training)

- Systems can become outdated quickly

- Lack of awareness among professionals

- Access and usage barriers

Use in Our Project:

1. Focus on complete communication:

Our system will not only support basic needs but also allow users to express emotions and social interactions.

2. Simplify the interface:

Based on the insight that usability is important, we will design a simple and easy-to-use interface with large icons and minimal steps.

3. Improve speed of communication:

We aim to reduce the number of steps required to form sentences, making communication faster

4. Support multiple input/output modes:

The system will include icons, text display, and speech output to improve communication effectiveness.

5. Reduce dependency on training:

The system will be intuitive so that users can operate it without extensive learning.

6. Make the system adaptable:

We will allow customization so that the system can adjust to different users and situations.